

Stacker News Network Analysis

Investigating the most rewarding behaviour and the social network dynamics within the forum Staker News

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Abstract

This paper deals with a pioneering platform called "Stacker News" that has caught the attention of researchers and enthusiasts alike for its unique intersection of content creation, social engagement, and cryptocurrency. At the heart of this intriguing online forum lies an innovative microtransaction system, where members can manifest their appreciation for content through the transfer of "sats" – abbreviation from "satoshis," the smallest unit of Bitcoin, representing $\frac{1}{100'000'000}$ of a bitcoin. This research explores the multifaceted interactions within Stacker News, aiming to understand how this unique blend of social engagement and microtransactions shapes user behavior and content value. Through rigorous data scraping using Python and advanced statistical analysis in R, the paper dissects the complex web of user interactions, content creation patterns, and digital value exchange. The research navigates the intricate dynamics of the Stacker News network, uncovering how economic incentives influence both social connections and content popularity. This work delves into the strategies that drive user engagement and reward accumulation, examining the roles of content posting, commenting, and other platform-specific activities. The findings highlight the impact of the platform's microtransaction system on content visibility and community interactions, offering novel insights into the emergent behaviors at the confluence of digital social networks and cryptocurrency economics.

Introduction

This paper embarks on a comprehensive exploration of the Stacker News forum, founded in June 2021 by the computer scientist Keyan Kousha. Stacker News is considered a direct heir of the well-known Bitcoin Talk forum, a discussion forum created by the Finnish developer Martti Malmi and still considered the main cryptocurrency-related online forum.

The *about* section of the Stacker News GitHub repository defines the forum as "It's like Hacker News but we pay you Bitcoin", portraying the internal bitcoin-based tipping system as one of the main features of the forum. With regards to this aspect, Stacker News presents some similarities with a small social discussion site called Witcoin (announced in December 2010 and defunct since 2011) in which users had

the ability to tip each other using bitcoin transactions¹.

Stacker News merges the classical features of online forums (5) (4) (7) (12) with the innovative proposal of rewarding users with bitcoin for their activities, posts and comments. Moreover, every user can express their appreciation for some others' content with bitcoin tips, transforming every user into a *de-facto* economically-driven forum member.

Payment system

Before deepening into the forum dynamics and features, it's crucial to provide a summary of how Stacker News technically enables users to tip each other in the platform.

Tips and payments between forum users are denominated in *satoshi* (*sats*), which are fractions of bitcoin, a cryptocurrency. Specifically, 1 satoshi is the smallest unit of a bitcoin, corresponding to $\frac{1}{100'000'000}$ of a bitcoin.

The bitcoin cryptocurrency is a byproduct of the Bitcoin protocol, a payment system created in 2008 by an unknown individual called Satoshi Nakamoto (8). Briefly, the Bitcoin protocol is a distributed *peer-to-peer* network of nodes (computers) whose goal is to prevent the *double spending problem*² by forcing every node to store the whole history of transactions occurred in the network. The protocol is usually referred to as *blockchain* (11), because the transactions occurred in the network are arranged into an irreversible layered registry in the form of a chained sequence of blocks, that are effectively batched transactions.

For technical reasons that go beyond this work, the Bitcoin network cannot handle a vast amount of transactions³. In order to deal with the throughput problem, a consistent percentage of bitcoin *micro-transactions* are processed using a technology called **Lightning Network (LN)** (9), that is

¹As of October 2023, the Witcoin website is down and scarce information can be found about it other than this entry in the BitcoinWiki

²The double-spending problem is specifically enhanced by the digital realm because of the easiness in copying information. Double spending happens when a digital token is reused for different concurrent transactions.

³This is usually referred to as "the transaction throughput scalability problem". Throughput is generally defined as the "rate at which data is transferred through a system". With regards to bitcoin transactions, throughput is defined as the "amount of transactions that can be processed by the network in a fixed period of time".

a layer-2 solution to increase the bitcoin transaction throughput.

Briefly, LN protocol is a network of nodes in which node pairs are connected by payment channels of a size agreed upon by the two peers. The whole LN can be considered a giant component through which payments flow based on the shortest path between the payer and the receiver⁴

Tips, rewards and economical transactions that occur inside the Stacker News forum use the LN protocol; since transactions occurring in the LN protocol are *off-chain* transactions (meaning that they are not included in the Bitcoin chain of blocks), there is no way to retrieve precisely the relationship between payers and receivers. Interestingly enough, the Stacker News reveals the aggregated amount of sats received by every single user in their profile page.

Paper structure

In order to unveil the network dynamics of this platform, the structure of this paper is divided into three main components: Methodology, Data Analysis and Network Analysis. A final conclusion section will serve as a summary of the paper findings.

The **Methodology** section will outline the research approach adopted for the analysis. This will encompass the data scraping methods, the variables under consideration, and the analytical techniques employed to derive meaningful insights from the data at disposal. The Methodology section serves as the foundation upon which the entire study is built, providing a clear roadmap for the reader to understand the intricacies of the research process.

The **Data Analysis** section reveals how Stacker News' rewards system generally impacts user behavior. The researchers corrected total sats received by users to account for 'jailed' users who manipulated their earnings. The study observes that creating content, commenting, forwarding sats to another user, and spending time on the forum are ways to earn sats both from tips and platform rewards. The analysis of post types shows that discussions are the most rewarded and boosting content visibility through sats is common.

The **Network Analysis** section involves examining the interactions within the Stacker News community. It considers user interactions as directed relationships and uses attributes like user type, sats received, and total stacked sats to understand user behavior and the forum's social dynamics. Moreover, this section presents a time based analysis, showing the growth and development of the Stacker News community over time. It highlights the centralization of the forum, the rise of big hubs, and the constant average degree of separation within the network.

This structured approach aims at providing readers with a well-organized and insightful journey through the forum analysis, offering a holistic view of the platform's network dynamics and its unique contribution to the broader digital landscape.

⁴The pathfinding algorithm depends on the software implementation used by the payer node. Based on the number of GitHub forks, the LND implementation seems to be the most used. LND uses Dijkstra's algorithm for payment pathfinding

Methodology

This section delineates the structured approach employed in the investigation of the most economically-rewarding behaviour inside the forum, focusing on the dynamic and multifaceted nature of data collection and analysis from the Stacker News forum. The methodology unfolds in two interconnected phases: web scraping and data analysis. The former encompasses the selection of the Stacker News forum, the utilization of Python with the `BeautifulSoup` library for web scraping, and the subsequent data filtering and cleaning processes. The latter phase involves the statistical analysis carried out using R, with a specific emphasis on temporal dynamics achieved through the segmentation of data into six-month periods.

Web-Scraping. The web scraping process was conducted on the Stacker News forum website using Python, specifically the `BeautifulSoup` and `Requests` library. Python was chosen for its versatility and robust ecosystem in web scraping tools. The first step of this section was understanding the structure and organization of the content in the forum: every element on the forum (posts and comments) is identified as an *item* and can be accessed with the URL `https://stacker.news/items/<item-number>`, where the *item-number* is a progressive integer. This structure allows to access every item by simply cycling from item number 1 to the last item number present in the forum (as of October 2023). The following analysis is based on the first 270'000 items of the forum.

The web-scraping activity produced three different datasets:

- **Posts table**, containing information about items post (scraped from the item page);
- **Comments table**, containing information about times comment (scraped from the item page);
- **Users table**, containing information about the user (scraped from the user profile page).

Data Analysis. The subsequent phase of data analysis unfolds within the R programming language, leveraging its statistical prowess and visualization capabilities. Within this stage, a nuanced approach was firstly adopted to unravel the nature and the structure of the economical rewards. This section will provide a general comprehension of the data collected with the previous phase and will give some insights about the users' behaviour inside the forum.

Network Analysis. With this section the researchers will analyse the features of the social network outlined by the forum, firstly by analysing the general graph (thus conducted on all the history of the forum), secondly by dividing into discrete six-month periods and extracting some insights with regards to trends, patterns, and fluctuations of network parameters over time.

Data analysis

Stacker News stands out as a pioneering platform that adds a compelling twist to the traditional exchange of ideas. Unlike conventional forums, Stacker News introduces a unique dimension by incorporating economic reward system, where users have the ability to financially appreciate content by donating satoshis.

The first aspect enlightened by this research regards a correction of the total sats received by the users. Analysing the scraped data it is clear that there are some high-earnings users (according to the cumulative stacked amount computed using the *comments* and *posts* datasets). Checking the total amount of sats from *users* dataset, the measures of the same feature are different. This is due to the fact that these are 'jailed' users which faked their total earnings by sending sats from an account to another account. This fraudulent actions were used as sponsorship because, during the first month of the forum, the visibility of a content was defined by considering the amount of Sats received by the content posted. Luckily, this trick was revealed and administrators changed the post indexing system. As of now, the *visibility ratio* of a post is calculated by considering both a trust index (given by the founder of the forum *k00b* to every user) and the number of unique users that tipped the post (thus, the indexing of items is not strictly linked to the sats amount received)⁵. This fraudulent users are still forum members, therefore they still should be included in the research. However it is important for further analysis to mention that by executing the validation process between the two stacked amounts available is possible to isolate these users and is also possible to isolate the users that received consistent amounts via forwarding or in form of platform rewards.

Another bias which needs to be explained is the fact that users can earn sats by 'forwarding' actions. In other words, the user X creates a posts and decides that a custom percentage of the earnings from that post has to be forwarded to the Y user. These amounts are clearly included in the total of sats earned by the post, but they aren't actually delivered to the post creator but rather to the user that receives the forwarding. Thus, these amounts can be captured by looking at the difference between the stacked amount by users from the *users* table and the resulting from the cumulative computation of the other datasets.

To answer the research question and to investigate the most rewarding behaviour it is possible to affirm that the best option is to analyse and include the total amount of Sats computed on the *Posts* and *Comments* datasets. That is because the researcher are interested on the behaviour linked to posting and commenting, but also interested in the rewards received.

With regards to the tipping system, it's crucial to clearly highlight all the possible revenues streams that a user has at dispose in the forum:

- Posting and Commenting: creating content or comments on others' posts unlocks the possibility of receiving tips to that specific content;

⁵More information about the Web of Trust indexing are retrievable at this link

- Forwarding: by creating a new post item, any user can choose to forward a custom percentage of sats received by the post to one or multiple other users;
- Rewards (internally called "zaps"): every 24 hours the forum automatically gives a reward to every user, based on the last 24 hours behaviour and for spending time on the website.

Posts Posting on Stacker News is different from other on-line platform. In fact, users are given the chance to select the type of posts they are willing to create. More specifically the post typologies are:

- **Discussion**, that is a plain text item that enabling the user to create a lengthy post. This post typology supports comments and replies;
- **Link**, that restricts the content submission to a single URL;
- **Poll**, which enables the user to create a poll about a topic;
- **Bounty**, that allows users to specify a task and a payout prize that the user is willing to pay for the task being completed by anyone.

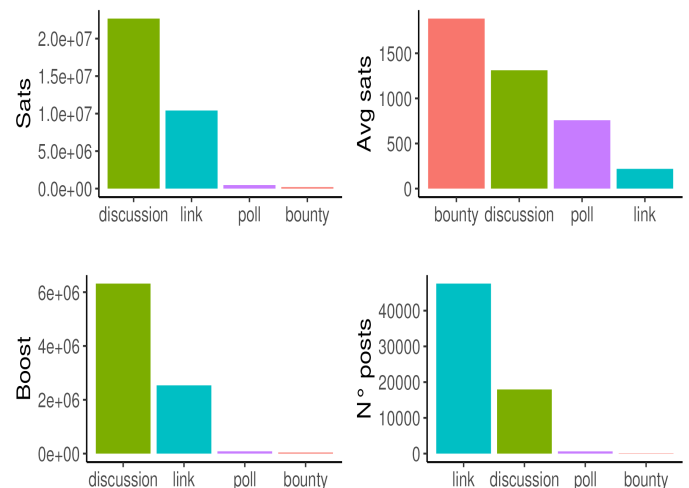


Figure 1: Post statistics by type

Looking at Figure 1 it is clear that the most rewarded post-type considering the sum of Sats is *discussion* while the majority of the post items are links. This difference is due to the boosting effect: during the post creation process the user has the possibility to boost his content using 25k sats. This is very similar to the up-vote mechanism on *Reddit* where posts with more up-votes are more visible than others. It can be simply understood by considering boosts as sponsorships. The leading position of discussion posts in this ranking can be addressed to boosting effect. Checking the average sats,

the leading position is occupied by bounty. This can be addressed to the small number of this typology which create a sort of bias computing the mean.

It is clearly not possible to answer the research question using just the information about Post Types, but this image gives some hints about discussion-type which collects more rewards considering both sum and mean.

Author	Sats received	N°Post	Boost	Sats per post	First post date
siggy47	3542513	586	100000	6045	2022-05-24 02:27:35
k00b	2822539	2055	217620	1373	2021-06-11 19:26:02
02838203b0	1500000	1	0	1500000	2022-08-17 08:17:47
scarcity	1000000	2	0	500000	2022-05-19 19:43:26
Rsync25	565793	901	0	628	2023-02-02 10:59:05
spain	546706	5	17	109341	2021-07-15 09:39:35
birdseye21	464890	241	0	1929	2023-06-14 16:59:47
sn	456038	167	2	2731	2021-07-15 23:03:48
OpenSats	451000	1	0	451000	2022-08-26 19:24:43
cryptocoin	408218	2666	1349	153	2021-12-18 15:49:03

Table 1: Posters ranking

Table 1 shows the top 10 users considering the cumulative sum of Sats received from tips on posts and comments. Firstly, users *02838203b0*, *scarcity*, *spain*, *OpenSats* are distinguished by the fact that they have created a very small amount of posts, but they have earned very large amount of Sats. It must be remarked that, while *Opensats* and *spain* seem to be legit users ⁶, the earnings of *02838203b0* and *scarcity* come only from anomalous posts and have been recognised by the administrators as fraudulent users. This is due to the early computation of the visibility ratio and the fraudulent behaviour of the users which sent Sats reciprocally from an account to another to increase their visibility. In order to accomplish this research, these users need to be considered as outliers because they do not give insights to answer the research question and to spot the most rewarding behaviours. Regarding the user named *K00b*, it is important to mention that he is the creator of the forum and the fact that he occupies on of the top positions of the ranking can be due to the fact that he was the first to post and he creates lots of content during the first months of the forum to enlarge the dimension and discussion on Stacker News. Moreover, he is still active and recently he has started to use boost to sponsor his content. Another interesting aspect enlightened by Table 1 regards the fact that boosts are not essential to earn Sats. Certainly, having visibility on contents is a great opportunity, but considering the cost of a sponsorship (25k sats), focusing on the quality of the content is more rewarding. Considering the number of posts created by every users and the date of the first post, it is possible to confirm the general idea that being active on the forum will make users collect more rewards.

⁶*Opensats* is confirmed to be the account of a not-for-profit association that raised funds also through the platform, whereas *spain* received an anomalous amount of donations in a link post, but seems to be an active average user according to the comment pattern.

Comments Similarly to other social media platforms, Stacker News enables users to submit comments. Visitors and users can manifest appreciation to a comment by donating sats to the commentor.

Author	Sats received	N° comments	Sats per comment
tech5	2106857	122	17269.31967
ekzyis	2009846	3291	610.70981
random	1289081	694	1857.46542
utxoclub	1187049	13	91311.46154
k00b	1055987	7735	136.52062
anarkio	1038800	300	3462.66667
DarthCoin	721010	9299	77.53629
justsmile	507483	87	5833.13793
tldr	506048	84	6024.38095
nemo	435487	3339	130.42438

Table 2: Commentors ranking

The reasoning for users with a small number of comments is similar to what was said for poster. Looking to Table 2 it is also important to remember that users named *K00b* and *ekysis* are administrators of the forum and their behaviour could be biased by their obvious interest in creating high volume of content traffic on the forum.

The conclusions of this section will deal with the relationships between Number of Post/ Number of Comments and Sats received. This will test the hypothesis which can be summarised as “The more the content (post or comment), the higher the earnings”.

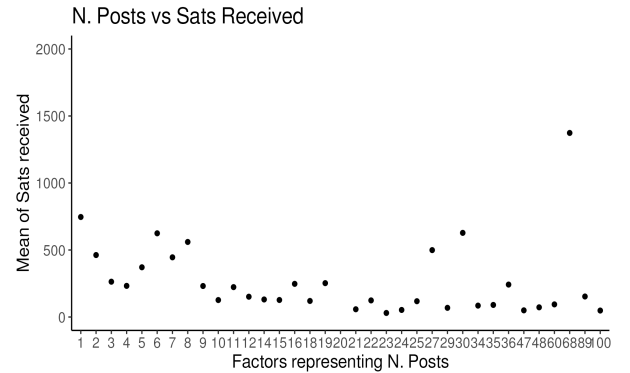


Figure 2: Posts vs Sats received

In Figure 2 the x-axis represents the Number of Posts considering 100 breaks on the ordered variable *NPosts* which create 100 groups containing customers with the same or similar amount of posts. The y-axis represents the mean of the Sats Received per group. It is possible to affirm that there is not a relationship between number of posts and rewards. Posting lots of content is not a key element to understand the most rewarding behaviour on Stacker News.

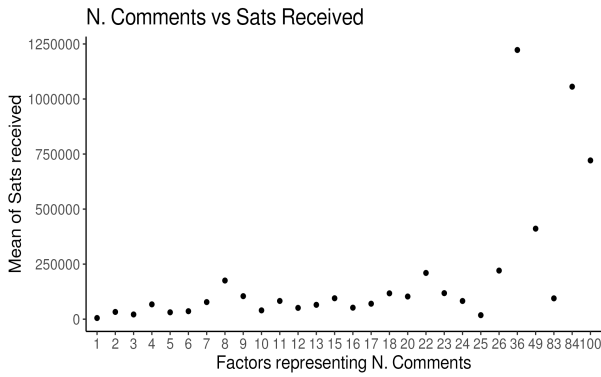


Figure 3: Comments vs Sats received

The graph in Figure 3 is constructed similarly to the graph in Figure 2, but with the x-axis representing the number of comments. In this scenario, there seem to be a relationships between the number of comments and the rewards collected by the user. It is important to notice that x-axis does not shows all the possible groups, this is due to the fact that the distribution of the variable *Ncomments* contains more observations with a low number of comments. However, it seems to partially confirm the hypothesis whereby users producing lots of comments tend to earn more Sats.

Network Analysis

Stacker News is an intricate forum with regards to user interaction and the creation of relationships between users, moreover it gives the opportunity to observe the behaviours of users in an economic-reward based forum platform. The role of **network analysis** in such an environment is to outline the phenomena that cannot be observed using a classic data-oriented analysis, providing a clearer picture of the network of users, their connections and interactions and therefore a better understanding of the forum platform (10).

Network setup

In order to setup the network, the crawled data has to be organized with respect to the following distinction:

- data referring to users (vertex attributes):
- data referring to user interactions (edge attributes).

Edges Because of the nature of the forum, the relationship between users is considered to be a **directed relationship** whereby the direction of interactions is determined as a

Comment $\longrightarrow$ Post

directed relationship. Thus, the edge A - B represents the situation in which the user A commented an item of user B, being it a post item or a comment item. Given this structure, the weight of the connection between two users is computed as an integer that describes the number of the directed connections between the users (3).

Vertex Users are the vertex (nodes) of the graph. Their attributes are determined as follows:

- **Type.** Type attribute signals the role of the user in the forum, therefore this attribute can assume the value *contributor* if the user is identified as a project contributor to the *stacker.news* repository, otherwise the attribute has the value *other*.
- **Sats.** This attribute contains an integer representing the total number of satoshis that the user received from posts and comments. This attribute is different from the *Totstacked* attribute.
- **Totstacked.** This attribute contains an integer representing the total number of satoshis the user received, based on the user profile data.
- **Reward.** This attribute contains an integer that is the difference between *Totstacked* and *Sats*.

The attributes *Sats*, *Totstacked* and *Reward* are all referring to the amount that the user earned using the platform, but they are computed with different techniques. The attribute *Sats* is computed as the sum of satoshis received from the user in posts and comments that she/he created. This value therefore encapsulates the **aggregated economical value that the other forum users decided to assign to the specific user as for his/her forum interactions**. As remarked in Data analysis section, during the initial months of the forum some users exploited the tipping system in order to make their posts appear in the initial positions of the homepage. These users were then identified and the ranking system of the platform changed. Nevertheless, the amount of satoshis received by these users cannot be modified.

For this reason a new attribute is included in the graph (*Totstacked*), which is the total amount stacked by the user as indicated in the user profile. The importance of this attribute is related to the fact that the forum enables users to forward a custom percentage of the satoshis earned to some other users, therefore the amount of satoshis received by the post could be different from the amount that is actually delivered to the post creator. Moreover, the proactive behaviour of users is rewarded by the forum through a *daily reward* gift. These forum features create the need for a measure that can encapsulate the net amount of satoshis, including forwarded satoshis and rewards.

The *rewards* attribute captures the difference between these two values. A positive reward amount means that the user received rewards and/or forwarded amounts, meaning that somehow the forum users or the platform itself rewarded him/her for his/her behaviour. On the other hand, a negative amount means clearly that the user decided to forward some satoshis earned with a post, signaling for his/her collaborative and appreciative behaviour with respect to other users.

General graph

In this paper the wording 'General graph' indicates the weighted directed graph with the following characteristics:

- **users** are represented as nodes;
- **user interactions** are represented as edges;

- **number of interactions** between a specific pair of users are represented as edge weights;
- the graph is the result of the aggregated forum data, from june 2021 to october 2023 .

The graph is of size $|V| = 5'551$ and order $|E| = 94'339$, thus indicating a **sparse graph**. Indeed the theoretical maximum of edges for this directed graph is computed as:

$$|E|_{max} = 2 \times \binom{|V|}{2} = |V|(|V| - 1) = 30'808'050$$

This consideration strengthens the theoretical assumption (and the empirical evidence) whereby *real networks* tend to be sparse and not dense, thus:

$$|E| \ll |E|_{max}$$

Since the analysis considers the relationships between users as directed relationships, the analysis of nodes degree must be carried on differentiating between **total degree**, **inner degree** and **outer degree**. In this context the in-degree represents the number of interactions that the user *received*, thus the number of comments that the user received on the items that he/she created. The out-degree represents the number of outgoing links from the node, thus the number of comments that the user submitted to items created by other users.

Table 3 shows the top 10 users with the higher total degree.

User	Total Degree	In-Degree	Out-Degree
k00b	2702	1330	1372
DarthCoin	2635	1139	1496
nym	2336	228	2108
siggy47	1982	860	1122
kr	1421	890	531
cryptocoin	1407	690	717
ekzyis	1303	609	694
nerd2ninja	1171	505	666
kepford	1145	345	800
nout	1116	546	570

Table 3: Author Statistics

This ranking is confirmed by analysing the in-degree and out-degree sorted list, with a two significative exceptions. The users *sn* and *saloon* appear in the in-degree top 10 despite the critically low amounts of outgoing connections, 41 and 1 respectively. A further analysis of user profiles clarified that both users are reserved bots created by the platform administrators in order to host specific forum threads and weekly livestreams.

The total degree distribution confirms what is generally observed in real networks (2). Specifically, there is a substantial difference between the the average total degree and the median total degree, with the former equal to 34 and the latter equal to 5, highlighting the crucial role of hubs in the forum balance.

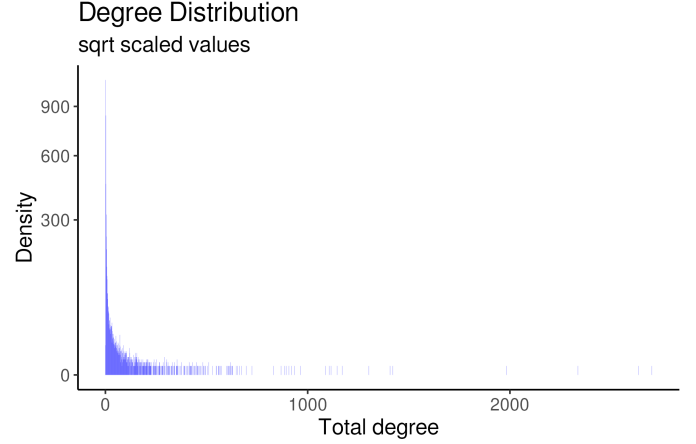


Figure 4: Total degree distribution

Forum contributors Given the nature of the forum and its strong roots in the open-source community, the list of current contributors to the forum development can be easily extrapolated from the analysis of the commits and contributions to the forum development on official GitHub repository. Moreover, the active certified contributors to the forum are included in the contributors.txt file. For this narrow subset of users the relationship between the in-degree and the out-degree is linear; moreover the most rewarded contributors were the ones with an higher total degree.

Degree and rewards The observation of nodes degree is functional to the economical-reward analysis because of the general assumption that can be formulated, whereby the more the interactions inside the forum, the more the users tend to be rewarded.

As observed for the contributors group, most users tend to distribute on the secant straight line, meaning that generally the amount of outgoing and incoming interactions (in the form of out-degree and in-degree) is similar for the majority of the users. Significantly enough, most high earning users tend to be in the group of high total-degree users, with 13 users that received more than 1 million sats ($\approx \text{€}340$ as of november 2023) and 3 of them being forum contributors ⁷.

Figure 5 shows users that earned more than 10k sats ($\approx \text{€}4,2$ as of november 2023); these users are the top 13% users of the forum with regards to earnings. The color of the bubbles is based on the total amount earned (according to the user profile page) and the ones labeled are the users that earned more than 1mln sats. Further verifications

⁷Note that the sats amounts here are measured with the *tot-stacked* parameter.

confirmed that the users *scarcity* and *02838203b0* are jailed users that exploited the reward system in the early days of the forum (as stated previously). Given this clarification, the graph clearly shows that the majority of high earning users belong to the group whose members registered more than 10 in both the in-degree and out-degree. Notably, only 19.8% of the users scored +10 in both out-degree and in-degree. Moreover, an analysis of the rewards outlined that users with an high total degree tend to receive high rewards (from the platform and the other users), with the exception of the user *siggy47*, that received a negative net amount of rewards ⁸.

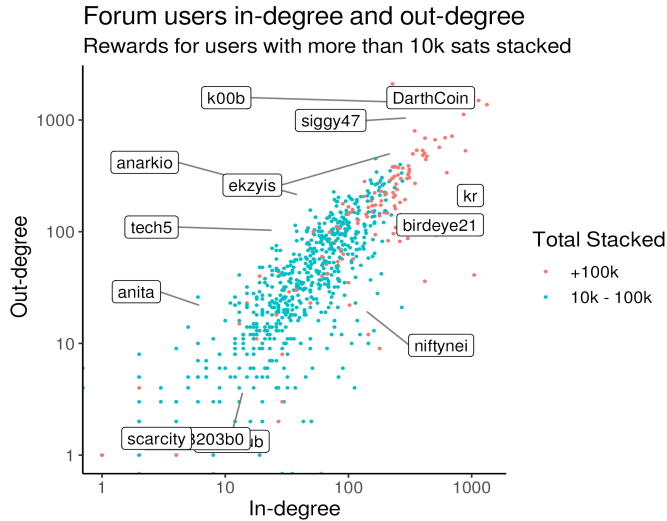


Figure 5: Forum users in-degree and out-degree: rewards for users with more than 10k sats stacked

The previous considerations confirm the initial general hypothesis: a proactive behaviour inside the forum - in the form of commenting (out-degree) and receiving comments (in-degree) - usually generates a positive economical reward in the form of economical tips paid by the other users and by the forum itself.

Earnings per post ratio In order to better address the research question, new variables are introduced in the analysis, focusing on the posting patterns and the tendencies in content creation. A qualitative analysis of the forum highlights the nature of posts with *link* category, that clearly are used to share external content (as remarked earlier); on the other hand the other post categories (*discussion*, *poll* and *bounty*) are designed to enhance and stimulate the internal forum activity. From the data available in the *posts* and *comments* tables is possible to extract the following ratios:

- **PoTo** (percentage of posts on the total items created), that is a proxy of the 'original content' created by the user;

⁸This user fits into the category of behaviour outlined previously whereby he/she persistently forwarded his/her post tips to others, therefore the amount sats received by his/her posts is way higher than the amount that actually was delivered to the user account.

- **PoLi** (percentage of link items on the total posts created), that is a proxy of 'external referencing content' created by the user.

These measures confirmed the obvious linear correlation between the number of items created and the total degree, meaning that the more content the user creates, the more people he/she is getting in touch with. With regards to the relationship between the number of items created and the total earnings, the two variables don't seem to be linearly correlated. Moreover, nor the **PoTo** nor the **PoLi** ratios show any correlation with the total earnings of the user.

In order to extrapolate meaningful insights from the data, the **earnings-to-post ratio (etp)** is derived as:

$$etp = \sqrt{totstacked \times (1 - normalized\ total\ items)}$$

with:

- *totstacked* being the total earned by the user (retrieved from profile data);
- $(1 - normalized\ total\ items)$ being inverse of the normalized value (min-max normalization) of total items created by the user. In other words, the less the items created by the user, the higher the value.

The interaction between the two variables weights the earnings of the user according to his/her commitment to posting often. Given a similar *totstacked* value, the user that posted less items obtains an higher *etp* ratio.

The following graph shows the interaction between the total degree and the percentage of productive posts (measured as the inverse of the **PoLi**). The darker the point, the higher the **etp** value.

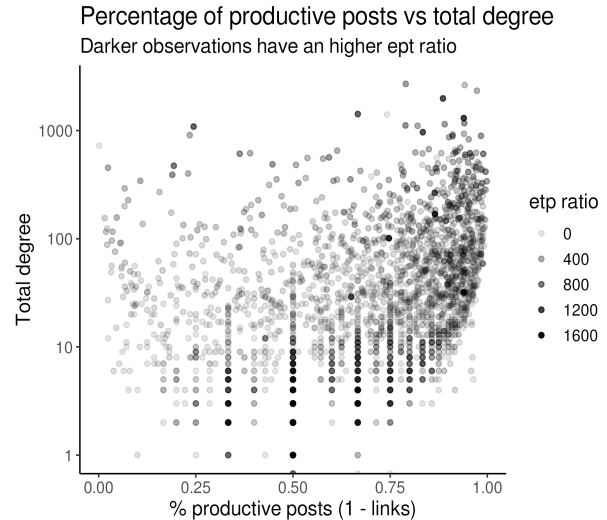


Figure 6: Percentage of productive posts vs total degree (darker observations have an higher earnings-per-post ratio)

Figure 6 shows that posting an high percentage of productive posts (that are discussions, bounties and polls) leads

to an higher total degree. Moreover, productive posts tend to obtain an higher **etp** ratio.

In conclusion, based on the past interactions and post history of users of stacker news, it emerges that creating original content in the form of discussions, polls and bounties is more likely to generate more tips and therefore more economical value. On the other hand, link-driven content (link items) tend to generate less interactions with other users and therefore less tips and rewards in the form of satoshis.

Network characteristics The quantitative and graphical analysis of the forum outlined the presence of 13 different components. Most forum users are connected with each other and part of the giant component, counting 5'538 out of 5'551 users. The other 12 components are *single-member* components with the exception of one, that includes the users *0307d4a092* and *020b3d342e*.

As occurring for most of the real world networks, Stacker News confirms the tendency of users to agglomerate into a giant network component.

Given the network's nature as a directed weighted network, the longest shortest path is calculated between the nodes *psztorc* and *svrnhdl*, resulting in a diameter of 24 edges.

Furthermore, in addition to the aforementioned details, the average distance between members within the giant component is just 3 edges. This finding aligns with the observations made in the 1990 Small World experiment (6) and was further explored in several other research works (1).

Periods breakdown

Overall, the relevance of the *rewards* and *totstacked* attributes is inevitably related to the general graph, since the *totstacked* attribute is retrieved from the user profile (that represents aggregated result). For this reason, the analysis involving the single periods is carried on based on the *sats* attribute as a proxy for the quantity earned by the user.

The table at the end of the chapter summarizes the results for the periodical analysis.

It should be remarked that the data are not cumulative but isolated data for the given period.

The data highlight how the Stacker News has experienced an incredible growth since its birth, particularly during the first and secon period, in which unique users grew by three-folds. With regards to users, it should also be remarked that the ones monitored and registered during the data gathering are uniquely the active users, the ones that submitted at least one item during the given period.

One of the most meaningful paramethers to observe the growth of a real network is constituted by the comparison between the actual number of edges and the theoretical maximum number of edges achievable (according to the number of nodes). It should be noted that this measure, described as a ratio between the two numbers, constantly decreased throughout the life of the forum, consolidating at 0.5% if considered the aggregated data. This behaviour indicates the obvious centralization of the forum and the rise of big hubs, which is confirmed also by the growth of the average total

degree and the contextual rise of a giant component containing most of the forum users.

In conclusion, the temporal analysis not only reaffirms the findings of the renowned Small World experiment but also provides additional evidence that the average degree of separation in the network remains consistently at 3.

	First	Second	Third	Fourth	Fifth	General
Total vertex	326	838	2'212	2'425	1'951	5'551
Total edges	1'692	8'331	32'447	29'710	30'962	94'339
A/T edges ratio	1.60%	1.19%	0.66%	0.51%	0.81%	0.31%
Mean total degree	10.38	19.88	29.34	24.5	31.74	33.98
Median total degree	3	4	5	5	6	5
N° of contributors	3	3	5	6	7	7
Components	4	8	10	11	8	13
Largest component	323	829	2'201	2'414	1'944	2'238
Diameter	13	16	14	15	26	24
Avg degree of separation	3.22	3.03	2.97	2.99	2.91	2.97

Table 4: Periods statistics

Conclusions

In conclusion, this investigation into Stacker News has revealed the profound impact of integrating cryptocurrency into a social media platform. The use of satoshis as a medium for content appreciation not only incentivizes user engagement but also fosters a unique ecosystem where social interactions are intertwined with economic rewards.

It has been observed that the platform's unique economic model encourages a culture of quality content creation and active participation. Discussions, polls, and bounties, as forms of original content, were found to be more lucrative in attracting tips, indicating a preference for substantial textual contributions over mere link sharing. This insight is crucial for platforms considering similar integrations of digital currencies.

Furthermore, network analysis highlighted the intricate relationships among users, revealing a community structure where proactive engagement correlates with economic benefits. The study of Stacker News contributes to the ongoing discourse on the integration of financial mechanisms in digital communities, showcasing how economic incentives can significantly influence online interactions and content creation.

The findings also prompt further exploration into the broader implications of such integrations. As digital platforms increasingly adopt cryptocurrency elements, understanding their effects on user behavior and community dynamics becomes essential. This study serves as a stepping stone for future research in this domain, encouraging exploration into how digital currencies can reshape social platforms, fostering new forms of interaction, collaboration, and value creation, including data about transactions between user which are unfortunately available for this research.

In essence, Stacker News stands as a pioneering example of the synergy between social media and cryptocurrency, offering valuable lessons for the evolution of digital commu-

nities. As we continue to navigate this intersection, the insights gained from this study will be instrumental in shaping the future landscape of online social and economic interactions.

Contributions

This work was redacted by Bersan Alberto and Tolin Filippo. The workload was distributed as follows:

- Data scraping (Bersan, Tolin)
- Data cleaning (Bersan)
- Data analysis and exploration (Bersan)
- Network analysis (Tolin)
- Paper (Bersan, Tolin)
- Power point presentation (Bersan, Tolin)

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